Amath SOW

Linkoping-Sweden| linkedin.com/in/amathsow | (+46) 734778756 | amath.sow@liu.se

EXPERIENCE

OlliTrip **Remote**
Machine Learning engineer *April 2022- June 2024*

- Built an AI based travel recommendation engine
- Built recommendation system for hotel using kafka, spark and elastic search
- Implemented a deep learning based attraction recommendation system using Restricted Boltzmann Machine
- Implement data pipeline and ML pipeline using Apache Airflow
- Deployed the models in AWS (Sagemaker, EC2, S3, etc)

Fellowship.ai **Remote**
Machine Learning Fellow *October 2022- January 2023*

- Worked on competitor analysis using Machine learning
- scraped nike products shoe
- Built data cleaning pipeline for csv data and image data loader for images shoes
- Built model pipeline using CLIP
- Implemented text and image embedding to fit into the CLIP model
- Used cosine similarity metric to get similar products
- Built a frontend application using streamlite and displayed the result.

Intelligent Robot Learning Lab **Alberta/Canada**
Reinforcement Learning researcher *June 2020 - January 2022*

- Implemented four Human in Loop Reinforcement Learning(HCRL) algorithms using tensorflow framework : TAMER (Training and Agent Manually via Evaluative Reinforcement), COACH (Convergent Actor Critic by Human Feedback), DEEP-TAMER (TAMER using Deep neural network), DEEP-COACH (COACH using Deep neural network)
- Built a new HCLR algorithm, Variance-Reduce COACH(VR-COACH) which interprets the human feedback as reward and apply variance reduction technique on policy gradient commonly used in RL
- Experimented with VR-COACH in the classic MountainCar environment and DEEP VR-COACH in Goal navigation task(Malmo Minecraft).

Huawei Technologies(3G/4G) **Senegal**
Radio access network/Application and service engineer *January 2015- September 2018*

- Involved in Radio Access Network (RAN) maintenance and monitoring for 3G and 4G.
- Involved in Application and Service (AS) maintenance for the network intelligent: CRM, CBS, API SMS, USSD, Network KPI monitoring.

PROJECTS

Jengo AI: AI-driven Property Maintenance Automation

- Developed LLM based agents to interpret tenant requests and classify issues.
- Built predictive models to suggest optimal vendor assignment and maintenance prioritization.
- Integrated AI workflows into the Jengo platform, enabling end-to-end automation of maintenance processes.
- Deployed AI services on AWS EC2 instances using Flask and Gunicorn for scalable, production-ready APIs.

- Implemented performance optimization techniques, including model fine-tuning and inference acceleration, to improve response times and accuracy.
- Collaborated with frontend developers to connect AI outputs with the centralized operations dashboard, enhancing usability for property managers.

Fashion AI : print generation using AI and 3D rendering

- Used latent diffusion pretrained model to generate print based on prompt
- Build react application that consume the latent diffusion model
- Build 3D rendering model using google model viewer
- Deployed the react application on AWS amplify
- Deployed the diffusion model on an AWS ec2 deep learning based instance using flask and gunicorn
- Implemented super resolution model to get better results.

Variance Reduced Convergent Actor Critic by Human Feedback(VR-COACH)

- Built Episodic COACH by modifying COACH algorithm and computed the policy gradient for each timestep but the gradient update is done at the end of the episode.
- Observed that the gradient update is similar to the one in REINFORCE except that we have the human feedback in place of the reward and the eligibility decay λ in place of the discount factor.
- Built a new algorithm VR-COACH, which is an actor-critic based algorithm in which the human feedback is interpreted as reward.
- Experimented with VR-COACH in the classic MountainCar environment and demonstrated that it learns faster than COACH and TAMER.
- Upgraded our original VR-COACH to his DEEP version (DEEP VR-COACH) where agent's policy is represented as a deep neural network and apply Convolutional Auto Encoder strategy, a Feedback Replay Buffer and entropy regularization in order to learn complex task in a reasonable time
- Demonstrated the effectiveness of DEEP VR-COACH in the rich Malmo Mine-craft environment while comparing it with Deep COACH and DEEP TAMER.

EDUCATION

Reasoning and Learning Lab(ReaL) — **Sweden**
PhD student — *December 2023*

African Master in Machine Intelligence(AMMI) — **Ghana**
Master in Machine Intelligence — *March 2021*

Gaston Berger University — **Senegal**
Electronic and Telecommunication engineer — *November 2014*

Gaston Berger University — **Senegal**
Bachelor in Maths, Physics and computer science — *August 2011*

Planned Graduation Date: November 2027

Papers

Free Space Optical Channel Estimation Based on Deep Learning Algorithms — FOAN Conference 2023

Master thesis: Deep Reinforcement Learning from Variance-Reduced Policy-Dependent Human Feedback (https://github.com/amathsow/personal/blob/main/ammi_thesis_amath_sow.pdf) — March 2021